

1.4 Sequences & Series

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

A *sequence* is an ordered list of numbers, each one called a *term*. A *series* is the sum produced by adding those terms together. The study of sequences and series lies at the centre of pure mathematics: arithmetic and geometric progressions appear in compound interest, population models, and physics problems; sigma notation compresses large sums into a single symbol; the binomial theorem extends polynomial expansion from exponents like $(1+x)^5$ into the infinite world of $(1+x)^{1/2}$ or $(1+x)^{-3}$. This topic also introduces the idea of a *limit* — the value a sequence approaches as more and more terms are produced — which foreshadows calculus.

1. Sequences — Notation and Definition

A sequence is a function whose domain is the positive integers (or sometimes $\{0, 1, 2, \dots\}$). The n th term is denoted a_n or u_n ; the whole sequence is written $(a_n)_{n \geq 1}$ or simply $a_1, a_2, a_3, \dots$

Sequence. An ordered list of numbers $a_1, a_2, a_3, \dots$ indexed by a positive integer n . Each number a_n is a *term*; n is the *position* or *index* of that term. A sequence may be finite (with a last term) or infinite (continuing indefinitely).

Two Ways to Specify a Sequence

There are two standard ways to describe a sequence:

- Position-to-term formula** — an explicit expression $a_n = f(n)$ giving the n th term directly.
Example: $a_n = 3n + 2$ gives 5, 8, 11, 14, ...
- Term-to-term (recurrence) formula** — a rule $a_{n+1} = g(a_n)$ expressing each term in terms of the preceding one(s), together with an initial term. Example: $a_{n+1} = a_n + 3$ with $a_1 = 5$ gives the same sequence 5, 8, 11, 14, ...

Recurrence relation. An equation $a_{n+1} = g(a_n)$ (or involving more earlier terms) that expresses each term of a sequence as a function of one or more preceding terms. Together with initial condition(s) this uniquely determines the sequence.

Exam tip — deducing a formula from the first few terms. If a sequence begins 2, 5, 8, 11, 14, ... and you spot a common difference $d = 3$, the explicit formula is $a_n = a_1 + (n - 1)d = 2 + 3(n - 1) = 3n - 1$. Always verify your guess on at least two terms (e.g. $n = 1$ gives 2 ✓; $n = 4$ gives 11 ✓). Do not just check the first term — many different formulae agree at $n = 1$.

Worked Example 1 — Converting between position-to-term and term-to-term

A sequence is defined by $a_{n+1} = 2a_n - 1$ with $a_1 = 2$. Compute $a_1, \dots, a_5$ and find a position-to-term formula.

1. Compute terms:
2. $a_1 = 2$
3. $a_2 = 2 \cdot 2 - 1 = 3$
4. $a_3 = 2 \cdot 3 - 1 = 5$
5. $a_4 = 2 \cdot 5 - 1 = 9$
6. $a_5 = 2 \cdot 9 - 1 = 17$.
7. Differences 1, 2, 4, 8 double $\rightarrow$ suggests 2^n involvement.
8. Try $a_n = 2^{(n-1)} + 1$:
9. $n=1$: $1 + 1 = 2$ ✓
10. $n=2$: $2 + 1 = 3$ ✓
11. $n=3$: $4 + 1 = 5$ ✓
12. $n=4$: $8 + 1 = 9$ ✓
13. $\therefore a_n = 2^{(n-1)} + 1$. // verified explicit formula

2. Arithmetic Progressions (AP)

An arithmetic progression (AP) is a sequence in which each term is obtained from the previous one by adding a fixed constant, called the *common difference*.

Arithmetic progression. A sequence $a_1, a_2, a_3, \dots$ such that $a_{n+1} - a_n = d$ for all n , where d is a fixed real constant (the *common difference*). Equivalently, the n th term is $a_n = a + (n-1)d$, where $a = a_1$ is the first term.

Sum of an AP

Let $S_n = a_1 + a_2 + \dots + a_n$ denote the sum of the first n terms. Writing the sum forwards and backwards and adding term-by-term gives Gauss's formula:

$$S_n = \frac{n}{2} [2a + (n-1)d] = \frac{n}{2}(a + l),$$

where $l = a_n = a + (n-1)d$ is the last (or n th) term. The second form is especially useful when both endpoints of the sum are known.

Exam tip — which AP sum formula to use. Use $S_n = \frac{n}{2}(a + l)$ when you already know the first and last term of the sum (e.g. "sum all integers from 1 to 100"). Use $S_n = \frac{n}{2}[2a + (n-1)d]$ when you know the first term and the common difference but not the last term. Both give the same answer — pick whichever avoids extra arithmetic.

Worked Example 2 — Sum of the first 100 positive integers

Compute $1 + 2 + 3 + \dots + 100$.

1. This is an AP: $a = 1, d = 1, n = 100, l = 100$.
2. Using $S_n = n/2 \cdot (a + l)$:
3. $S_{100} = 100/2 \cdot (1 + 100) = 50 \cdot 101 = 5050$.
4. Check via other formula:
5. $S_{100} = 100/2 \cdot [2 \cdot 1 + 99 \cdot 1] = 50 \cdot 101 = 5050 \checkmark$.

// Gauss famously produced this answer as a schoolboy by pairing
 // 1+100, 2+99, ..., 50+51 — fifty pairs each summing to 101.

Worked Example 3 — AP with known first and last terms

An AP has 20 terms, first term 5 and last term 100. Find the common difference and the sum.

1. Last term: $a_n = a + (n - 1)d$.
2. $100 = 5 + 19d \Rightarrow 19d = 95 \Rightarrow d = 5$.
3. Sum: $S_n = n/2 \cdot (a + l)$
4. $S_{20} = 20/2 \cdot (5 + 100) = 10 \cdot 105 = 1050$.
5. Quick sanity check – average term $\times$ number of terms:
6. $\text{mean} = (5 + 100)/2 = 52.5; 52.5 \times 20 = 1050 \checkmark$.

Applications of APs

- **Salary with fixed annual raise.** A worker earns £25 000 in year 1 with a £1 000 annual increase. Lifetime earnings over 30 years form an AP.
- **Pile of logs / cans.** A triangular stack with 1 log on top, 2 in the next row, $\dots$, n at the base has total $\frac{n}{2}(1 + n)$.
- **Walking / driving with constant change.** First day walk 5 km, each day walk 0.5 km more — total distance after n days is an AP sum.

3. Geometric Progressions (GP)

A geometric progression (GP) is a sequence in which each term is obtained by multiplying the previous one by a fixed non-zero constant called the *common ratio*.

Geometric progression. A sequence $a_1, a_2, a_3, \dots$ such that $a_{n+1}/a_n = r$ for all n , where $r \neq 0$ is a fixed real constant (the *common ratio*). The n th term is $a_n = a \cdot r^{n-1}$, where $a = a_1$.

Sum of a GP

Let $S_n = a + ar + ar^2 + \dots + ar^{n-1}$. Multiplying by r gives $rS_n = ar + ar^2 + \dots + ar^n$.

Subtracting:

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1.$$

The two forms are algebraically identical; the first is neater when $|r| > 1$, the second when $|r| < 1$.

Sum to Infinity — Convergence

If $|r| < 1$ then $r^n \rightarrow 0$ as $n \rightarrow \infty$, and the infinite sum has a finite value:

$$S_\infty = \frac{a}{1 - r}, \quad \text{valid only if } |r| < 1.$$

If $|r| \geq 1$ the partial sums do not settle to any single number — we say the series *diverges* and S_∞ is undefined.

Convergence of a GP. An infinite geometric series $a + ar + ar^2 + \dots$ converges if and only if $|r| < 1$. When it converges, its sum is $S_\infty = a/(1 - r)$. When $|r| \geq 1$ (and $a \neq 0$) it diverges — the partial sums either grow without bound or oscillate.

Limit of a sequence. The sequence (a_n) has limit L (written $a_n \rightarrow L$ or $\lim_{n \rightarrow \infty} a_n = L$) if, informally, a_n can be made arbitrarily close to L by taking n sufficiently large. If no such L exists, the sequence diverges.

Exam tip — check convergence *before* summing. Every year OCR scripts lose marks by blindly applying $S_\infty = a/(1 - r)$ without verifying $|r| < 1$. If the ratio is $r = 2$ or $r = -3$, the infinite sum is undefined — writing down $a/(1 - r)$ anyway will score zero on that part. Always state the validity condition explicitly in your answer.

AP vs GP — at a glance

Feature	Arithmetic progression	Geometric progression
Recurrence	$a_{n+1} = a_n + d$	$a_{n+1} = r \cdot a_n$
n th term	$a + (n - 1)d$	$a \cdot r^{n-1}$

Sum to n terms	$\frac{n}{2}[2a + (n - 1)d]$	$a(r^n - 1)/(r - 1)$
Sum to infinity	Always diverges (unless all terms zero)	Converges iff $ r < 1$, with $S_\infty = a/(1 - r)$
Test if given	Successive differences constant	Successive ratios constant
Example	3, 7, 11, 15, ...	3, 6, 12, 24, ...

Worked Example 4 — GP with known sum

A GP has 5 terms with first term 2 and sum 62. Find the common ratio ($r > 0$).

1. $S_n = a(r^n - 1)/(r - 1)$:
2. $62 = 2(r^5 - 1)/(r - 1)$.
3. $\Rightarrow 31 = (r^5 - 1)/(r - 1) = 1 + r + r^2 + r^3 + r^4$.
4. Try $r = 2$: $1 + 2 + 4 + 8 + 16 = 31$ ✓.
5. $\therefore r = 2$. // common ratio
6. Verify series: 2, 4, 8, 16, 32. Sum = 62 ✓.

Worked Example 5 — Sum to infinity

A convergent GP has sum to infinity 8 and first term 6. Find the common ratio, and write down the first four terms.

1. $S_\infty = a/(1 - r)$:
2. $8 = 6/(1 - r) \Rightarrow 1 - r = 6/8 = 3/4 \Rightarrow r = 1/4$.
3. Check convergence: $|r| = 1/4 < 1$ ✓ — valid.
4. First four terms: 6, $6 \cdot (1/4) = 3/2$, $3/2 \cdot (1/4) = 3/8$, $3/8 \cdot (1/4) = 3/32$.
5. Sum check: $6 + 1.5 + 0.375 + 0.09375 + \dots \approx 8$ ✓.

Applications of GPs

- **Compound interest.** £1000 invested at 5% per year grows as $1000, 1000(1.05), 1000(1.05)^2, \dots$ — a GP with $r = 1.05$.
- **Radioactive decay.** Mass remaining after n half-lives is $M_0, M_0/2, M_0/4, \dots$ — a GP with $r = 1/2$.
- **Repeating decimals as fractions.** $0.\overline{3} = 0.3 + 0.03 + 0.003 + \dots = \frac{0.3}{1-0.1} = \frac{1}{3}$.
- **Bouncing ball.** A ball dropped from h metres rebounds to a fraction r of its previous height, so total distance travelled is a converging GP sum.

4. Sigma Notation

Sigma notation compresses long sums into a compact symbol. The Greek capital sigma Σ means "add up".

$$\sum_{k=1}^n f(k) = f(1) + f(2) + f(3) + \dots + f(n).$$

Here k is the *dummy index* (any letter will do — i, j, r are common alternatives). The lower limit is the starting value; the upper limit is the last value (inclusive). Changing the letter does not change the sum.

Standard Sums

Sum	Closed form	Used for
$\sum_{k=1}^n 1$	n	Counting
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$	AP sum (natural numbers)
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$	Sum of squares — used in Pearson correlation, moment of inertia
$\sum_{k=1}^n k^3$	$\left[\frac{n(n+1)}{2} \right]^2$	Sum of cubes — equals $(\sum k)^2$, a surprising identity

Linearity of Sigma

Sums distribute over addition and scalar multiplication:

$$\sum_{k=1}^n [\alpha f(k) + \beta g(k)] = \alpha \sum_{k=1}^n f(k) + \beta \sum_{k=1}^n g(k).$$

This is used to split a complicated sum into pieces you can evaluate using standard results.

Changing Limits

To compute $\sum_{k=m}^n$ where $m > 1$, subtract:

$$\sum_{k=m}^n f(k) = \sum_{k=1}^n f(k) - \sum_{k=1}^{m-1} f(k).$$

Worked Example 6 — Using standard sum results

Evaluate $\sum_{k=1}^{20} (3k^2 - 2k + 1)$.

1. Split using linearity:
2. $\sum (3k^2 - 2k + 1) = 3 \sum k^2 - 2 \sum k + \sum 1$.
3. Apply standard results with $n = 20$:
4. $\sum k^2 = 20 \cdot 21 \cdot 41 / 6 = 2870$.
5. $\sum k = 20 \cdot 21 / 2 = 210$.
6. $\sum 1 = 20$.
7. Combine:
8. $3(2870) - 2(210) + 20$
9. $= 8610 - 420 + 20$
10. $= 8210$.
11. $\therefore \sum_{k=1}^{20} (3k^2 - 2k + 1) = 8210$.

Exam tip — watch the index start. $\sum_{k=0}^n$ has $n + 1$ terms, not n . $\sum_{k=1}^n$ has n terms. A frequent slip is taking $\sum_{k=0}^4 ar^k$ to have 4 terms (the upper limit) and using $n = 4$ in the GP sum formula,

when it actually has *five* terms ($k = 0, 1, 2, 3, 4$), so $n = 5$. Count the terms explicitly before plugging in.

5. Binomial Expansion — Positive Integer Index

The binomial theorem gives an expansion of $(1 + x)^n$ when n is a non-negative integer. The coefficients are the *binomial coefficients*, appearing in Pascal's triangle.

Binomial coefficient. For non-negative integers n and k with $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, read " n choose k ". It counts the number of ways to choose k objects from n distinguishable objects and is always a positive integer.

Binomial Theorem (Positive Integer n)

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

For a general binomial $(a + b)^n$:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

For a positive integer n the expansion has exactly $n + 1$ terms and *terminates*. No convergence condition is needed: the expansion is a finite polynomial identity, true for all real x .

Pascal's Triangle (first six rows)

n	Row of coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$	Sum of row = 2^n
0	1	1
1	1, 1	2
2	1, 2, 1	4

3	1, 3, 3, 1	8
4	1, 4, 6, 4, 1	16
5	1, 5, 10, 10, 5, 1	32
6	1, 6, 15, 20, 15, 6, 1	64

Each entry is the sum of the two entries diagonally above: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ — this is *Pascal's rule*.

Exam tip — finding a specific term. In $(a + b)^n$ the term containing b^k is $\binom{n}{k}a^{n-k}b^k$. For example, the x^3 term in $(1 + 2x)^6$ comes from $k = 3$: coefficient = $\binom{6}{3} \cdot 1^3 \cdot 2^3 = 20 \cdot 8 = 160$, so the term is $160x^3$. You rarely need the whole expansion — just isolate the term you want.

Worked Example 7 — Binomial expansion of $(1 + 2x)^6$

Expand $(1 + 2x)^6$ up to and including the term in x^3 .

1. Use $(1 + u)^6 = 1 + 6u + 15u^2 + 20u^3 + \dots$ with $u = 2x$.
2. Term-by-term:
3. 1
4. $6 \cdot (2x) = 12x$
5. $15 \cdot (2x)^2 = 15 \cdot 4x^2 = 60x^2$
6. $20 \cdot (2x)^3 = 20 \cdot 8x^3 = 160x^3$.
7. $\therefore (1 + 2x)^6 = 1 + 12x + 60x^2 + 160x^3 + \dots$ (terms in x^4, x^5, x^6 omitted)

// Full expansion (for interest, not required):
 // $1 + 12x + 60x^2 + 160x^3 + 240x^4 + 192x^5 + 64x^6$.

Worked Example 8 — Specific term in $(3 - x)^8$

Find the coefficient of x^4 in the expansion of $(3 - x)^8$.

1. General term: $C(8, k) \cdot 3^{(8-k)} \cdot (-x)^k = C(8, k) \cdot 3^{(8-k)} \cdot (-1)^k \cdot$
2. Set $k = 4$:
3. $\text{coef} = C(8, 4) \cdot 3^4 \cdot (-1)^4$
4. $= 70 \cdot 81 \cdot 1$
5. $= 5670.$
6. $\therefore$ coefficient of x^4 is 5670.

6. Binomial Expansion — Rational / Negative Index

For a real (not necessarily integer) exponent n , the binomial series still has a meaning, but as an *infinite* series that only converges for $|x| < 1$.

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots, \quad |x| < 1.$$

Notice two differences from the integer case:

- The expansion is infinite — it does not terminate because $n(n-1)(n-2)\cdots(n-k+1)$ is never zero when n is not a non-negative integer.
- It is only valid for $|x| < 1$. Outside this range the series diverges and gives nonsense.

Generalised binomial coefficient. For any real n and non-negative integer k , $\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$. When n is a non-negative integer and $k > n$ one of the factors in the numerator is zero, so $\binom{n}{k} = 0$, recovering the terminating case.

Expanding $(a + bx)^n$

When the constant inside the brackets is not 1, factor it out first:

$$(a + bx)^n = a^n \left(1 + \frac{bx}{a}\right)^n, \quad \text{valid for } \left|\frac{bx}{a}\right| < 1, \text{ i.e. } |x| < |a/b|.$$

Then expand $(1 + u)^n$ with $u = bx/a$ and multiply through by a^n .

Validity Range Summary

Form	Expansion	Valid for
$(1 + x)^n$, n positive integer	Finite polynomial ($n + 1$ terms)	All real x
$(1 + x)^n$, n negative or fractional	Infinite series	$ x < 1$
$(a + bx)^n$, n negative or fractional	$a^n(1 + bx/a)^n$, infinite series	$ bx/a < 1$, i.e. $ x < a/b $
$(1 - x)^n$, n negative or fractional	Replace x with $-x$ in standard expansion	$ x < 1$

Exam tip — always state the validity range. OCR mark schemes award an explicit mark (often one of three or four) for stating the correct range of x for which the expansion is valid. For $(1 + 3x)^{-2}$, valid for $|3x| < 1$, i.e. $|x| < 1/3$. Writing just "valid for $|x| < 1$ " is wrong here — the coefficient inside matters.

Worked Example 9 — First four terms of $(1 + 3x)^{-2}$

Find the first four terms of the expansion of $(1 + 3x)^{-2}$ in ascending powers of x , and state the range of x for which it is valid.

- General expansion:
- $(1 + u)^n = 1 + nu + \frac{n(n-1)}{2!} \cdot u^2 + \frac{n(n-1)(n-2)}{3!} \cdot u^3 + \dots$
- Here $n = -2$, $u = 3x$.
- Compute coefficients:
- $n = -2$
- $\frac{n(n-1)}{2!} = \frac{(-2)(-3)}{2} = 3$
- $\frac{n(n-1)(n-2)}{3!} = \frac{(-2)(-3)(-4)}{6} = -4$.
- Terms:
- 1
- $\frac{(-2)(3x)}{1} = -6x$
- $3 \cdot (3x)^2 = 3 \cdot 9x^2 = 27x^2$
- $\frac{(-4) \cdot (3x)^3}{1} = -4 \cdot 27x^3 = -108x^3$.
- $\therefore (1 + 3x)^{-2} = 1 - 6x + 27x^2 - 108x^3 + \dots$

14. Validity: $|3x| < 1 \Rightarrow |x| < 1/3.$

7. Approximation Using the Binomial

The binomial series lets us approximate powers of numbers close to 1. If x is small the first few terms of $(1 + x)^n$ give a good estimate.

Worked Example 10 — Approximating $\sqrt{1.02}$

Use a binomial expansion up to the x^2 term to approximate $\sqrt{1.02}$, and estimate the error.

1. Write $\sqrt{1.02} = (1 + 0.02)^{(1/2)}$.
 2. Expansion with $n = 1/2$, $u = 0.02$:
 3. $(1 + u)^{(1/2)} = 1 + (1/2)u + (1/2)(-1/2)/2! \cdot u^2 + \dots$
 4. $= 1 + u/2 - u^2/8 + \dots$
 5. Substitute $u = 0.02$:
 6. $\approx 1 + 0.01 - (0.0004)/8$
 7. $= 1 + 0.01 - 0.00005$
 8. $= 1.00995.$
 9. Calculator value: $\sqrt{1.02} = 1.00995049\dots$
 10. Error $\approx 5 \times 10^{-7}$ — extremely accurate because u is small.
- // The next term (u^3 term) has coefficient $(1/2)(-1/2)(-3/2)/6 = 1/16$,
 // giving $u^3/16 \approx (8 \times 10^{-6})/16 = 5 \times 10^{-7}$, consistent with observed error.

Exam tip — how many terms for a target accuracy? Each successive term in the binomial series shrinks roughly by a factor of x (when $|x|$ is small). So if you want four-decimal-place accuracy and $|x| = 0.02$, two terms past the constant usually suffice. Rule of thumb: the first omitted term is the dominant error — estimate it and check it is below your tolerance.

8. Recurrence Relations — Convergence to a Limit

Some recurrence relations $a_{n+1} = g(a_n)$ produce sequences that settle down to a limit L . If $a_n \rightarrow L$ and g is continuous, then taking limits of both sides of $a_{n+1} = g(a_n)$ gives $L = g(L)$ — the limit must be a *fixed point* of g .

If $a_n \rightarrow L$ as $n \rightarrow \infty$, then $a_{n+1} \rightarrow L$ too, so $L = g(L)$.

Solving $L = g(L)$ locates the candidate limits. Not every fixed point is attracting; whether the sequence actually converges depends on the starting value and on $|g'(L)|$ (a detail beyond the specification), but candidate limits are found by this fixed-point equation.

Worked Example 11 — Convergent recurrence

Let $a_{n+1} = \frac{1}{2}(a_n + 4)$ with $a_1 = 0$. Compute $a_1, \dots, a_5$, and find the limit L .

1. Compute terms:
2. $a_1 = 0$
3. $a_2 = (0 + 4)/2 = 2$
4. $a_3 = (2 + 4)/2 = 3$
5. $a_4 = (3 + 4)/2 = 3.5$
6. $a_5 = (3.5 + 4)/2 = 3.75$
7. $a_6 = (3.75 + 4)/2 = 3.875$
8. Terms approach 4 from below.

// Fixed point equation: assume limit L exists and let $n \rightarrow \infty$.

9. $L = (L + 4)/2$

10. $2L = L + 4$

11. $L = 4.$ $\therefore a_n \rightarrow 4$ as $n \rightarrow \infty$.

// Distance to limit halves each step: $|a_n - 4| \rightarrow 4, 2, 1, 0.5, 0.25, \dots$

// geometric decrease with ratio $1/2$, confirming convergence.

Worked Example 12 — Recurrence giving an AP

A sequence satisfies $u_{n+1} = u_n + 5$ with $u_1 = 3$. Find a closed-form expression for u_n and hence u_{20} and $\sum_{k=1}^{20} u_k$.

1. Recurrence $u_{n+1} - u_n = 5$ identifies an AP with first term $a = 3$ and common difference $d = 5$.
2. Closed form: $u_n = a + (n - 1)d = 3 + 5(n - 1) = 5n - 2$.
3. Check: $u_1 = 3 \checkmark$, $u_2 = 8 \checkmark$, $u_3 = 13 \checkmark$.
4. $u_{20} = 5 \cdot 20 - 2 = 98$.
5. Sum of first 20 terms:
6. $S_{20} = 20/2 \cdot (u_1 + u_{20}) = 10 \cdot (3 + 98) = 10 \cdot 101 = 1010$.
7. Check with other formula:
8. $S_{20} = 20/2 \cdot [2 \cdot 3 + 19 \cdot 5] = 10 \cdot [6 + 95] = 10 \cdot 101 = 1010 \checkmark$.

9. Worked Exam-Style Problems — Combined Techniques

Worked Example 13 — Repeating decimal as a fraction

Express the recurring decimal $0.\overline{27} = 0.272727\dots$ as a fraction in lowest terms using an infinite GP.

1. Write as a sum:
2. $0.272727\dots = 0.27 + 0.0027 + 0.000027 + \dots$
3. This is a GP with:
4. first term $a = 0.27 = 27/100$
5. common ratio $r = 0.01 = 1/100$ (each term is $1/100$ of the previous).
6. $|r| = 0.01 < 1$, so the series converges.
7. Sum to infinity:
8. $S_\infty = a/(1 - r) = (27/100) / (99/100) = 27/99$.
9. Simplify: $\text{gcd}(27, 99) = 9$, so $27/99 = 3/11$.
10. $\therefore 0.272727\dots = 3/11$.
11. Check: $3 \div 11 = 0.272727\dots \checkmark$.

Worked Example 14 — Compound interest

£2000 is deposited at 4% compound interest per year. Find the value after 10 years and the total interest earned.

1. After n years balance = $2000 \cdot (1.04)^n$ — a GP with $a = 2000$, $r = 1.04$.
2. After 10 years:
3. $V = 2000 \cdot (1.04)^{10}$.
4. $(1.04)^{10}$ — expand as $(1 + 0.04)^{10}$ using binomial ($n = 10$, positive integer).
5. $= 1 + 10(0.04) + 45(0.04)^2 + 120(0.04)^3 + \dots \approx 1.48024$.
6. (Calculator value: 1.48024428...)
7. $V \approx 2000 \times 1.48024 \approx \text{£}2960.49$.
8. Interest earned $\approx 2960.49 - 2000 = \text{£}960.49$.

Worked Example 15 — Wage calculation (AP application)

An employee earns £22 000 in year 1 and receives a £900 annual pay rise. Find total earnings over a 25-year career.

1. AP with $a = 22000$, $d = 900$, $n = 25$.
2. Salary in year n : $a_n = 22000 + 900(n - 1)$.
3. Year 25 salary: $22000 + 900 \cdot 24 = 22000 + 21600 = \text{£}43600$.
4. Sum:
5. $S_{25} = 25/2 \cdot [2 \cdot 22000 + 24 \cdot 900]$
6. $= 25/2 \cdot [44000 + 21600]$
7. $= 25/2 \cdot 65600$
8. $= 25 \cdot 32800$
9. $= \text{£}820\,000$.

// Alternatively: $S_{25} = 25/2 \cdot (22000 + 43600) = 12.5 \cdot 65600 = 820000$. ✓

10. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Sequences & Series.

- **Identify the sequence type first.** Common difference $\Rightarrow$ AP; common ratio $\Rightarrow$ GP; neither $\Rightarrow$ use recurrence or general formulae.
- **AP:** $a_n = a + (n - 1)d$, $S_n = \frac{n}{2}(a + l) = \frac{n}{2}[2a + (n - 1)d]$.
- **GP:** $a_n = ar^{n-1}$, $S_n = a(r^n - 1)/(r - 1)$. **Sum to infinity requires** $|r| < 1$.
- **Sigma notation:** memorise Σk , Σk^2 , Σk^3 ; use linearity to split complicated sums.
- **Binomial (positive integer n):** finite expansion with $n + 1$ terms, valid for all real x ; coefficients from Pascal's triangle.
- **Binomial (rational/negative n):** infinite series, valid for $|x| < 1$; for $(a + bx)^n$ factor out a^n first and validity becomes $|x| < |a/b|$.
- **Recurrence limits:** assume limit L exists, substitute $a_{n+1} = a_n = L$, solve $L = g(L)$.
- **State validity conditions** explicitly for infinite GP and non-integer binomial — OCR awards a mark for this.

Typical OCR examiner complaints (from mark reports).

- Using $S_\infty = a/(1 - r)$ without checking or stating $|r| < 1$.
- Omitting the validity range $|x| < |a/b|$ for the expansion of $(a + bx)^n$.
- Arithmetic slips in $n(n - 1)(n - 2)/3!$ when n is negative — sign errors abound.
- Wrong number of terms in a sigma sum (index starts at 0 vs 1).
- Forgetting to factor a^n out of $(a + bx)^n$ before applying the series formula.
- Writing $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \dots$ while missing the $2!$ in the denominator of subsequent terms' factorials.
- Failing to verify a guessed explicit formula on multiple terms, not just the first.
- Mixing up AP and GP formulae under exam pressure — confirm the structure before reaching for a formula.

Formula snapshot (memorise).

- AP n th term: $a_n = a + (n - 1)d$.
- AP sum: $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$.

- GP n th term: $a_n = ar^{n-1}$.
- GP sum: $S_n = a(r^n - 1)/(r - 1)$, $r \neq 1$.
- GP sum to infinity: $S_\infty = a/(1 - r)$, $|r| < 1$.
- Standard sigma sums: $\Sigma k = n(n + 1)/2$, $\Sigma k^2 = n(n + 1)(2n + 1)/6$, $\Sigma k^3 = [n(n + 1)/2]^2$.
- Binomial (integer n): $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$.
- Binomial (general n): $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$, $|x| < 1$.
- Fixed point of recurrence: $L = g(L)$ if the limit exists.